# Matthias Takele

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Software Engineer

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## WORK EXPERIENCE

**Guild** / Software Engineer II — May 2025 - Present

- Launched a pharmacy education benefit serving 1,000+ learners, building conditional React/TypeScript enrollment forms and Rails integrations supporting nearly **$18M in approved benefits**.
- Fixed duplicate-funding race conditions in Rails/PostgreSQL with idempotency safeguards and recovery from audit history, covering 80K+ approvals totaling **$160M+**.
- Built a resumable Node.js migration pipeline to move legacy reimbursement data into new payments services, integrating PostgreSQL, DynamoDB, Snowflake, and GraphQL.
- Added configurable payouts for books, materials, and fees to a separate education benefits platform, extending Python/FastAPI calculations and React/TypeScript policy controls.

**Datadog** / Software Engineer I — Aug 2023 - May 2025

- Built Python ETL pipelines from Amazon S3 to PostgreSQL, validating incoming data and reducing query latency by 20%.
- Developed a Go migration layer to move usage data onto a contract-aware schema while preserving existing customer-facing API behavior.
- Automated dataset deprecation and cleanup with scheduled jobs and monitoring, reducing stale data by 40% and decreasing manual maintenance.

**Datadog** / Software Engineering Intern — Jun 2022 - Aug 2022

- Built a PostgreSQL migration tracker that caught schema discrepancies in containerized environments, reducing migration errors by 90%.
- Aggregated and validated approximately **200M historical records** for usage and billing graphs, improving retrieval times by 30%.

## PROJECT EXPERIENCE

**Albie** / albie-app.com — 2025 - Present

- Built and shipped an iOS app for collaborative photo albums, journaling, and messaging using React Native/Expo, TypeScript, and a Node.js/PostgreSQL backend.
- Added semantic photo search and Ask Albie, an on-device AI assistant using image-text embeddings, local LLMs, structured outputs, and response validation.
- Implemented resumable media uploads and offline synchronization with durable queues and idempotent APIs, using Redis, Cloudflare R2, and Socket.IO for shared albums.

**RecipeVerse** / Demo / GitHub — Feb 2025

- Built a recipe discovery app with React/TypeScript, TF-IDF search, CSV indexing, and Firebase authentication for saved recipes.

**Spotify Top Artists Trivia** / Demo / GitHub — Dec 2024 - Jan 2025

- Turned Spotify top artists into personalized trivia using the Spotify Web API, Flask, and Jinja templates; deployed on Vercel.

## TECHNICAL SKILLS

**Languages** Python, TypeScript, Go, Ruby, SQL

**Frameworks** React, React Native/Expo, Rails, FastAPI, Node.js, Express, GraphQL

**Data and Infrastructure** PostgreSQL, Redis, Snowflake, DynamoDB, AWS, Docker, Kubernetes

## EDUCATION

**Massachusetts Institute of Technology** — May 2023

B.S. in Computer Science and Engineering